QP CODE: 26900525

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION OCTOBER 2026
FIFTH SEMESTER
DISCIPLINE SPECIFIC CORE COURSE (DSC) - MG5DSCSTA300 - APPLIED REGRESSION ANALYSIS (2024 ADMISSION ONWARDS)

SCHEME FOR VALUATION

Duration: 2 hours **Maximum Marks: 70**

PART A
Short answer type questions
Answer any 10 questions. Each question carries 3 marks.

1. Any three violations such as non-linearity, heteroscedasticity, autocorrelation/serial correlation, non-normal errors, multicollinearity among predictors. -1 mark each
2. Any three uses/ roles -1 mark each
3. Definition/ Formula -3 marks
4. Regression Model -1 mark
   OLS estimator -2 marks
5. Multiple Regression Model -1 mark
   Any two roles -2 marks
6. Definition of Efficiency/ Condition for Efficiency -3 marks
7. Any two remedies -3 marks
8. Any two limitations of OLS in the presence of outliers -3 marks
9. Definition/Model -3 marks
10. Any two correct reasons -3 marks
11. Diagnosis meaning with example -3 marks
12. Definition/ Model of the fixed and random effects -3 marks

PART B
Short essay type questions
Answer any 4 questions. Each question carries 7 marks.

13. Any four applications in detail -7 marks
14. Model and Components -5 marks
    Interpretation -2 marks
15. Any three needs with any one model selection criterion -7 marks
16. Hypotheses, Test statistic, Degrees of freedom, Decision -7 marks
17. Original model, Transformation, GLS model and GLS estimator -7 marks
18. Definition of multicollinearity, Correlation matrix, diagnostics -7 marks

PART C

Essay type questions

Answer any 1 question. Each question carries 12 marks.

19. Parametric regression approach and example models -6 marks
    Non-Parametric regression approach and example models -6 marks
20. Any four methods with detailed explanation/example -12 marks